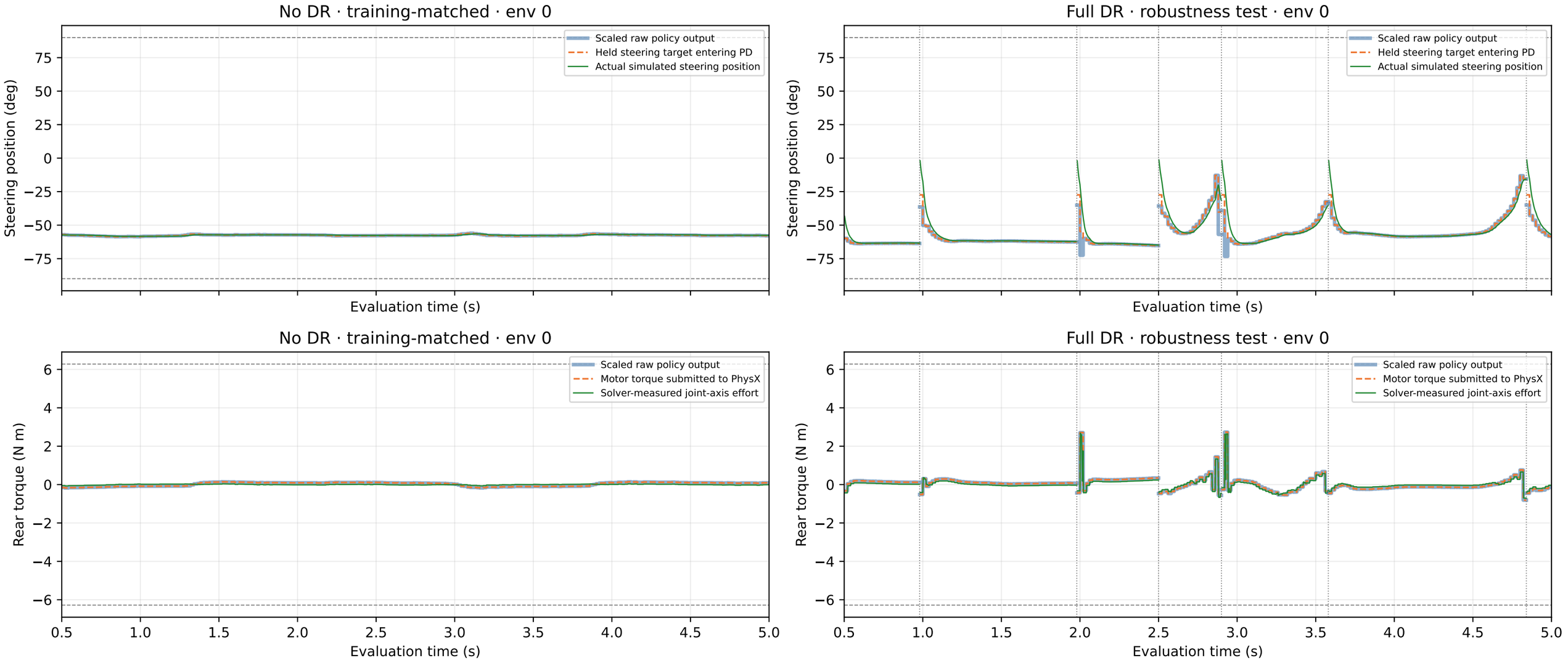
Isaac Lab · Trackstand-Balance-v1 · model_75300 · seed 42
No DR · training-matched · env 0
Full DR · robustness test · env 0
Scaled raw policy output
Held steering target entering PD
Actual simulated steering position
Steering position (deg)
75
50
25
0
−25
−50
−75
Evaluation time (s)
No DR · training-matched · env 0
Full DR · robustness test · env 0
Scaled raw policy output
Motor torque submitted to PhysX
Solver-measured joint-axis effort
Rear torque (N m)
6
4
2
0
−2
−4
−6
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Evaluation time (s)
50 Hz policy / 400 Hz physics · 0.5–5 s · resets break lines · no smoothing/time shift; linear axes